AI STUDENT SKILL GAP ANALYZER

Mini Project Documentation

Project Type: B.Tech CSE Mini Project

Development Platform: Base44 AI

Developer: Guru Prasanna

Live Project: https://weightless-skill-bridge-ai.base44.app

1. Project Overview

The AI Student Skill Gap Analyzer is an AI-powered web application designed to help students understand their current technical skills and identify the skills they need to improve for their desired career or job role. A student can enter their current skills and select a target job role. The application compares the student's existing skills with the skills required for the selected career, identifies skill gaps, and generates a personalized career roadmap.

2. Problem Statement

Many B.Tech and CSE students are unsure about which skills are required for their target job, which skills they already have, which skills they are missing, and what technologies they should learn next. As a result, students may spend time learning technologies that are not directly relevant to their career goals.

3. Proposed Solution

The proposed system provides a simple platform where students can enter or upload their skills, select a target job role, assess their current skill level, identify missing or weak skills, and receive a personalized learning roadmap.

4. Objectives

• Identify the skill gaps of students.
• Compare existing skills with career requirements.
• Provide personalized learning recommendations.
• Generate a structured career roadmap.
• Help students prepare for their desired job role.
• Make career planning easier for CSE students.

5. Target Users

• B.Tech students
• CSE and IT students
• Final-year students
• Fresh graduates
• Students preparing for placements
• Students planning their software careers

6. Key Features

Student Skill Input: Students can enter their current technical and professional skills.

Target Job Selection: Students can select roles such as Software Developer, Full Stack Developer, Frontend Developer, Backend Developer, Python Developer, Data Analyst, AI/ML Engineer, or Cloud Engineer.

Skill Gap Analysis: The application compares current skills with skills required for the selected job role.

Skill Gap Identification: The system highlights skills that need improvement and helps students prioritize them.

Personalized Roadmap: The application generates a step-by-step roadmap based on identified skill gaps.

Career Preparation: The roadmap helps students understand which technologies, concepts, and skills they should focus on.

7. System Workflow

Student
↓
Enter / Upload Skills
↓
Select Target Job Role
↓
Skill Assessment
↓
Compare Current Skills with Required Skills
↓
Identify Skill Gaps
↓
Prioritize Skills
↓
Generate Personalized Roadmap
↓
Student Learning & Progress

8. Technologies Used

Development Platform: Base44 AI
Application Type: Web Application
Main Concepts: Artificial Intelligence, Skill Gap Analysis, Career Recommendation, Personalized Learning Roadmap, User Profile Management, and Web Application Development.

9. Role of Base44 AI

Base44 AI was used to accelerate the development of the web application. It helped in creating the application structure, user interface, web pages, forms, dashboard components, application workflows, skill analysis functionality, and career roadmap interface. The project demonstrates how AI-assisted development platforms can help students convert an idea into a functional web application.

10. Advantages

• Easy to use.
• Student-focused.
• Saves career planning time.
• Identifies important skill gaps.
• Provides a structured learning roadmap.
• Helps students prepare for placements.
• Can be accessed through a web browser.
• Useful for CSE and IT students.

11. Future Enhancements

• AI-powered resume analysis
• Personalized course recommendations
• Coding practice recommendations
• Mock interview preparation
• Aptitude preparation
• Job vacancy matching
• LinkedIn profile analysis
• Skill certificate tracking
• Progress analytics
• AI career chatbot
• Real-time job market skill analysis

12. Project Link

Live Web Application:
https://weightless-skill-bridge-ai.base44.app

13. Conclusion

The AI Student Skill Gap Analyzer provides a practical solution for students who want to understand their career skill requirements. By comparing a student's current skills with the requirements of their target job role, the application identifies important skill gaps and generates a personalized roadmap. This project demonstrates how AI-assisted development using Base44 AI can be used to create practical and student-focused web applications.

14. Project Outcome

The final outcome is a functional web application that helps students answer three important questions:
What skills do I have?
What skills am I missing?
What should I learn next? The project can be used as a B.Tech mini project, portfolio project, hackathon

project, and GitHub project.

Project Information

Item	Details
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Application	Web Application
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